

Loop Equations and the Topological Phase of Multi-Cut Matrix Models

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Abstract

We study the double scaling limit of mKdV type, realized in the two-cut Hermitian matrix model. Building on the work of Periwal and Shevitz and of Nappi, we find an exact solution including all odd scaling operators, in terms of a hierarchy of flows of 2×2 matrices. We derive from it loop equations which can be expressed as Virasoro constraints on the partition function. We discover a “pure topological” phase of the theory in which all correlation functions are determined by recursion relations. We also examine macroscopic loop amplitudes, which suggest a relation to 2D gravity coupled to dense polymers.

1 Introduction

We study the double scaling limit of mKdV type, realized in the two-cut Hermitian matrix model. Building on the work of Periwal and Shevitz and of Nappi, we find an exact solution including all odd scaling operators, in terms of a hierarchy of flows of 2×2 matrices. We derive from it loop equations which can be expressed as Virasoro constraints on the partition function. We discover a “pure topological” phase of the theory in which all correlation functions are determined by recursion relations. We also examine macroscopic loop amplitudes, which suggest a relation to 2D gravity coupled to dense polymers. crnkovic@yalehepnet, or @yalehepnet, Address after Sept. 1991: CERN, Theory Division, CH-1211, Geneva 23, Switzerland. On leave of absence from Institute “Ruder Bošković,” Zagreb, Yugoslavia. mrd@ruhets.rutgers.edu moore@yalehepnet, or @yalehepnet

2 Method

where M , N , and P are matrices satisfying $M^2 = -N^2 = -P^2 = I$. Finally, redefining M we obtain (dropping tilde from M) $M = 4(D+q) - 3A$. We will need the resolvent $R_1(D+q) - 3 - 4$. In their work on the resolvents of matrix differential operators Gelfand and Dikii show that R satisfies

$$L(\theta) = \sum_{i=1}^n \ell(x_i, \theta) \quad (1)$$

Equation 1 is included to keep the sample paper-like while remaining small.

3 Conclusion

and that it has an asymptotic expansion $R(x, \theta) = \sum_{k=0}^{\infty} R_k \theta^{-k}$. From it is clear that, up to constants, the most general expansion of R

is $\sum_{k=0}^{\infty} R_k x^k$, where R_k is a matrix. Plugging $\sum_{k=0}^{\infty} R_k x^k$ into $\sum_{k=0}^{\infty} R_k x^k = \sum_{k=0}^{\infty} R_k x^k$ results in the recursion relations determining $R_0, R_1, R_2, \dots$. Setting $R_0 = I$, $R_1 = 0$, we obtain